2025 6th International Conference on Communications, Information, Electronic and Energy Systems (CIEES) | 979-8-3315-0119-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/CIEES66347.2025.11300092

# Short-Term Energy Demand Forecasting in a University Campus Using Machine Learning

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***Abstract*—The accurate forecasting of energy consumption is a cornerstone of efficient energy management, particularly within complex entities such as university campuses. These environments are characterized by non-stationary demand patterns susceptible to abrupt changes, a phenomenon known as Concept Drift, which poses a substantial challenge to conventional forecasting models. This paper introduces and validates a holistic, three-phase methodological framework designed to address these challenges. We conduct a rigorous comparative analysis of three machine learning algorithms—Random Forest (RF), LightGBM, and Long Short-Term Memory (LSTM)—using high-resolution temporal data from a university building. The framework's initial phase establishes a baseline and diagnoses the severe impact of Concept Drift. The second phase implements a dynamic sliding window retraining protocol, demonstrating its necessity for model adaptation. The final, decisive phase integrates advanced feature engineering, which proves to be the most critical component for achieving high-fidelity predictions. Our results indicate that this final optimization stage reduces the Mean Absolute Error (MAE) for all models to a convergence point of approximately 1.1 kWh. Notably, this enhancement elevates the performance of the LightGBM model to a level commensurate with the specialized LSTM network. The study concludes that the development of a robust methodological framework, encompassing drift adaptation and sophisticated feature engineering, is of greater importance than the selection of a specific algorithm in isolation.**

***Keywords—short-term load forecasting, machine learning, concept drift, feature engineering, energy management, sliding window, university campus.***

## I. Introduction

The global imperative for sustainable energy practices and the proliferation of Smart Grid technologies have intensified the need for accurate Short-Term Load Forecasting (STLF) [1]. STLF serves as a foundational component for numerous critical operations, including unit commitment, economic dispatch, and demand-side management [2]. Within this context, university campuses emerge as particularly compelling and complex subjects of study. Functioning as veritable micro-cities, they exhibit highly dynamic and heterogeneous energy consumption patterns, driven by a confluence of factors such as academic schedules, research activities with high-energy equipment, variable occupancy, and seasonal HVAC (Heating, Ventilation, and Air Conditioning) loads [3].

This inherent complexity makes campus environments prone to non-stationarity in their data-generating processes. Such temporal shifts in the underlying data distribution are formally known as Concept Drift [4]. When a predictive model is trained on historical data, it implicitly learns the statistical relationships of that specific period. If these relationships change over time—due to operational modifications, equipment upgrades, or behavioral shifts—the statically trained model becomes progressively obsolete, leading to a significant degradation in predictive accuracy. This challenge is a primary obstacle to deploying reliable, long-term forecasting solutions in real-world settings.

While the application of Machine Learning (ML) to STLF is well-documented [5], a substantial portion of the existing literature evaluates algorithms under the implicit assumption of data stationarity. Comparatively fewer studies present a comprehensive, end-to-end methodological framework that explicitly diagnoses and remedies the effects of Concept Drift in a real-world scenario.

## II. Literature Review

The field of STLF has evolved significantly, transitioning from classical statistical methods to more sophisticated machine learning paradigms.

### A. Classical and Machine Learning Approaches

Traditional time series forecasting has been dominated by statistical methods such as ARIMA (Autoregressive Integrated Moving Average) and its seasonal variant, SARIMA [6]. While effective for linear and stationary time series, these models are often inadequate for capturing the complex, non-linear relationships inherent in energy consumption data.

The advent of Machine Learning has provided a powerful alternative. Models such as Support Vector Regression (SVR), Random Forest (RF) [7], and Gradient Boosting Machines (e.g., XGBoost, LightGBM) [8] have demonstrated superior performance by learning complex patterns directly from the data. More recently, Deep Learning models, particularly Recurrent Neural Networks (RNNs) and Long



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Short-Term Memory (LSTM) networks [9], have become the state-of-the-art for sequential data, owing to their ability to model long-term temporal dependencies explicitly.

### B. Addressing Concept Drift

The problem of Concept Drift is a well-recognized challenge in stream mining and time series analysis. An effective forecasting system must be adaptive. The most common and practical approach for this is periodic model retraining [4]. The sliding window method, where the model is retrained on a fixed-size window of the most recent data, is a robust strategy for this purpose. It allows the model to "forget" obsolete data while continuously learning from the current data distribution. Wang et al. (2018) effectively demonstrated the use of a sliding window to improve the adaptability of an LSTM model for university building data [10].

### C. The Primacy of Feature Engineering

Regardless of the model's complexity, its performance is fundamentally constrained by the quality of its input features. Feature engineering—the process of creating new, informative features from existing data—is widely acknowledged as a critical determinant of model performance [11]. In STLF, this includes:

- **Temporal Features:** Extracting calendar-based information (hour, day, month).
- **Lag Features:** Using past values of the time series as predictors to capture autocorrelation.
- **Rolling Statistics:** Calculating moving averages or standard deviations to represent recent trends and volatility.

Fekri et al. (2021) highlighted the decisive role of feature engineering when applying Gradient Boosting models to city-level load forecasting [12]. Our work combines these three pillars—robust ML algorithms, an adaptive learning framework, and sophisticated feature engineering—into a single, cohesive methodology.

This paper aims to bridge this gap by presenting a holistic, three-phase framework for developing a robust STLF model. Our contributions are manifold:

- We provide a clear empirical diagnosis and quantification of Concept Drift's impact on standard ML models using a real-world energy dataset.
- We validate the efficacy of a dynamic sliding window retraining protocol as an essential strategy for model adaptation.
- We demonstrate conclusively that sophisticated feature engineering is the most potent tool for achieving high-fidelity forecasts, capable of transforming general-purpose regression models into specialized forecasters that rival the performance of state-of-the-art deep learning architectures.
- We offer a complete case study that can serve as a practical, reproducible guide for researchers and energy managers facing similar forecasting challenges.

The remainder of this paper is structured as follows: Section 2 details the dataset and the methodological framework. Section 3 presents and analyzes the empirical results from each phase. Section 4 discusses the implications of these findings, and Section 5 concludes the paper with a summary and directions for future research.

## III. Methodology

The methodological framework of this study is designed to be systematic, reproducible, and transparent. It encompasses data acquisition and preprocessing, a sophisticated feature engineering protocol, a comparative algorithmic framework, and a multi-phase experimental design aimed at rigorously evaluating the proposed forecasting approach.

### A. Data Acquisition and Preprocessing.

The empirical analysis is grounded in a high-resolution temporal dataset sourced from the energy monitoring systems of a building complex at the Hellenic Mediterranean University (HMU) in Heraklion, Greece. The dataset comprises two primary variables recorded at an hourly frequency:

- **Energy Consumption:** The target variable, measured in kilowatt-hours (kWh).
- **Ambient Temperature:** The primary exogenous predictor, measured in degrees Celsius (°C).

The observation period spans 2.5 years, from June 1, 2020, to December 30, 2022, providing a rich repository of diverse seasonal and operational patterns.

### B. Exploratory Data Analysis and Problem Formulation.

A preliminary Exploratory Data Analysis (EDA) was conducted to understand the intrinsic properties of the time series. Visual inspection of the data, as depicted in Fig. 1, was instrumental in shaping the research questions and experimental design.

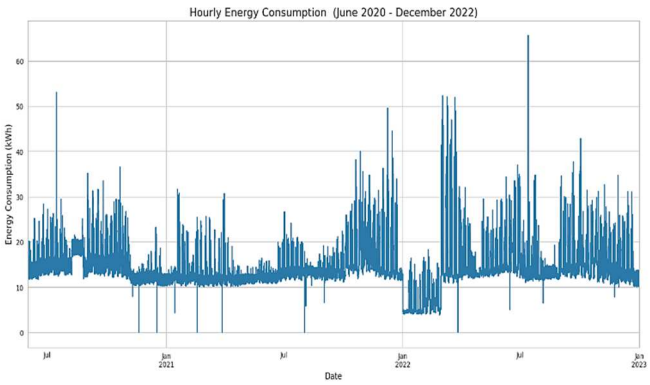



Fig. 1. Hourly energy consumption for the entire dataset period, clearly showing the concept drift in mid-2022.

The EDA revealed:

- **Multi-level Seasonality:** Pronounced and regular patterns corresponding to diurnal, weekly, and annual cycles.
- **Non-stationarity:** A clear structural break, or Concept Drift, was identified in the time series around mid-2022, characterized by a persistent increase in the baseline energy consumption. This observation was



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critical, as it invalidated the core assumption of stationarity required by many classical forecasting models and necessitated an adaptive modeling approach.

- **Exogenous Correlation:** A discernible relationship between temperature and energy consumption, particularly during summer months, highlighting the importance of including meteorological data.

*C. Data Cleansing and Imputation.*

To ensure data integrity and continuity, a two-step preprocessing protocol was applied. First, the time series was resampled to a fixed hourly frequency to account for any missing timestamps. Second, the resulting null values, which were sparse, were imputed using linear interpolation. This method was chosen over simpler techniques (e.g., forward-fill) as it provides a smoother and more realistic estimation of missing values in time series with discernible short-term trends.

*D. Feature Engineering Protocol*

Recognizing that the predictive power of any ML model is fundamentally constrained by the quality of its input data, we implemented a multi-faceted feature engineering protocol. The objective was to transform the raw time series into an enriched feature space that explicitly encodes temporal dynamics and domain-specific knowledge.

*E. Temporal Feature Extraction.*

The timestamp index was decomposed into a set of granular cyclical features to enable the models to learn calendar-based patterns:

- hour, day of week, month, quarter, year, day of year.

*F. Lag Feature Construction for Autocorrelation Modeling*

To provide the non-sequential models (RF and LightGBM) with a form of temporal memory, lag features were constructed. These features represent past values of the target variable and are crucial for modeling the strong autocorrelation present in energy data. We selected lags corresponding to key seasonal cycles:

- energy_lag_24h: Captures the diurnal pattern (consumption at the same hour on the previous day).
- energy_lag_168h: Captures the weekly pattern (consumption at the same hour on the same day of the previous week).

*G. Rolling Window Statistics for Trend and Volatility Capture.*

To equip the models with an understanding of the immediate trend and volatility, rolling window statistics were computed. These features act as a proxy for the first and second moments of the recent data distribution:

- rolling_mean_3h: The moving average over the last 3 hours, providing a smoothed indicator of the very recent consumption trend.
- rolling_std_24h: The moving standard deviation over the last 24 hours, quantifying the volatility or stability of the consumption pattern over the previous day.

*H. Exogenous Calendar Features*

To inject domain-specific knowledge into the model, a binary feature was created to identify public holidays in Greece.

- is_holiday: This feature is critical for capturing the abrupt and significant decrease in consumption that occurs during holidays, a pattern that cannot be inferred from temporal features alone.

*I. Algorithmic Framework*

Three distinct yet powerful ML algorithms were selected to ensure a robust comparative analysis.

- **Random Forest (RF):** A bagging-based ensemble method that constructs a multitude of decision trees on bootstrapped subsets of the data. It is known for its high accuracy and resistance to overfitting [7]. Implemented using Scikit-learn.
- **Light Gradient Boosting Machine (LightGBM):** A highly efficient implementation of the Gradient Boosting framework. It grows trees sequentially, with each new tree correcting the errors of the previous one, and employs novel techniques for speed and memory optimization [8]. Implemented using the official LightGBM library
- **Long Short-Term Memory (LSTM) Network:** A specialized type of Recurrent Neural Network (RNN) designed to overcome the vanishing gradient problem, making it exceptionally effective at learning long-range dependencies in sequential data [9]. The architecture employed consisted of a single LSTM layer (50 units), followed by a dense output layer. Implemented using TensorFlow/Keras

*J. Experimental Design and Validation Strategy.*

A three-phase experimental protocol was designed to systematically diagnose the challenges within the data and evaluate the incremental benefit of each proposed solution.

*K. Phase 1: Static Baseline for Concept Drift Diagnosis*

A conventional 80/20 chronological split was used. The models were trained on data from June 2020 to June 2022 and tested on data from July 2022 to December 2022. This phase utilized only the basic temporal features and was designed specifically to create a controlled environment where the effects of the mid-2022 Concept Drift could be isolated and quantified.

*L. Phase 2: Dynamic Adaptation via Sliding Window*

To address the issues identified in Phase 1, an adaptive learning strategy was implemented. The models were retrained for each month of the test period using a 12-month sliding window with a 1-month step size. The 12-month window size was deliberately chosen to ensure that the model always has access to a full annual cycle of data, allowing it to robustly learn seasonal patterns. This phase continued to use only basic temporal features to isolate the impact of the dynamic retraining strategy.

*Phase 3: Optimized Forecasting with Enriched Features*

The final phase represents the culmination of the methodology, combining the adaptive sliding window protocol from Phase 2 with the complete, enriched feature set from Section 3.2. This allowed for the evaluation of the



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synergistic effect of both adaptation and sophisticated feature representation.

### *M. Performance Evaluation Criterion*

The primary metric for evaluating model performance was the Mean Absolute Error (MAE). MAE was selected over the commonly used Root Mean Squared Error (RMSE) for two principal reasons [13]:

- **Interpretability:** MAE is expressed directly in the units of the target variable (kWh), making it easily interpretable as the average absolute forecast error.
- **Robustness:** MAE is less sensitive to large, sporadic errors (outliers) than RMSE, which penalizes large errors quadratically. In the context of energy forecasting, where unpredictable spikes can occur, MAE provides a more stable and representative measure of typical model performance.

$$MAE = \frac{\sum_{i=1}^{n}|y_i - x_i|}{n} = \frac{\sum_{i=1}^{n}|e_i|}{n} \quad (1)$$

## IV. Results and Analysis

The results from each phase demonstrate a clear and logical progression of performance improvement. A summary of the MAE scores for all phases is presented in Table I.

TABLE I. SUMMARY OF MAE (KWH) ACROSS ALL EXPERIMENTAL PHASES.

| *Model* | *Phase 1 MAE (Static)* | *Phase 2 MAE (Sliding Window)* | *Phase 3 MAE (Optimized)* |
|---|---|---|---|
| LSTM | **1.08** | 1.13 | **1.12** |
| LightGBM | 2.68 | 2.15 | **1.15** |
| RF | 3.00 | 2.28 | **1.18** |

### *A. Phase 1: Diagnosis of Concept Drift*

The results from the static training revealed a profound performance disparity. The tree-based models, RF and LightGBM, produced MAE scores nearly triple that of the LSTM. The visual evidence is unequivocal. The monthly aggregated predictions in Fig. 2 show a complete failure of the model to capture the correct magnitude of consumption.

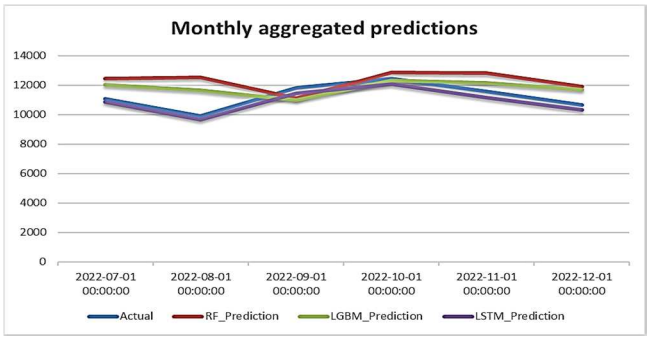



Fig. 2. Monthly aggregated predictions (Phase 1). The model systematically under-predicts consumption.

This systematic error is confirmed at the hourly level in Fig. 3, where the LightGBM model, while capturing the general diurnal shape of energy use, systematically fails on the magnitude. This demonstrates that the model, trained on pre-drift data, was applying obsolete patterns to the new, higher-consumption regime.

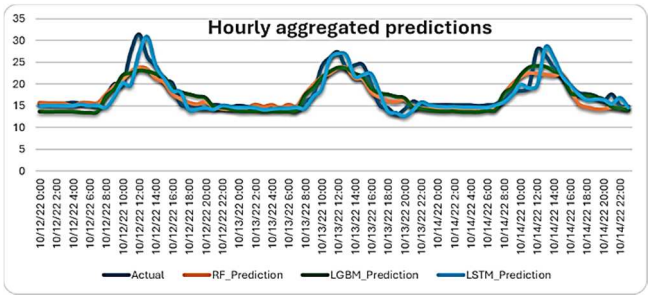



Fig. 3. Hourly predictions from the static LightGBM model (Phase 1). The model fails to adapt to the new energy baseline.

### *B. Phase 2: The Effect of Adaptation*

The implementation of the sliding window protocol yielded a significant error reduction for RF (~24%) and LightGBM (~20%). As shown by the monthly aggregated predictions in Fig. 4, this approach successfully corrected the systematic bias observed in Phase 1.

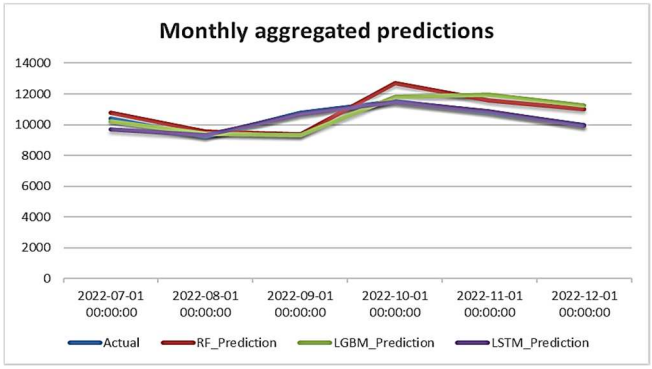



Fig. 4. Monthly aggregated predictions (Phase 2). The model now correctly captures the overall magnitude.

However, while necessary, this adaptation was not sufficient for high-precision forecasting. The weekly predictions in Fig. 5 show that the model, although in the correct range, still struggles to accurately track the weekly cycle, often missing the peaks.

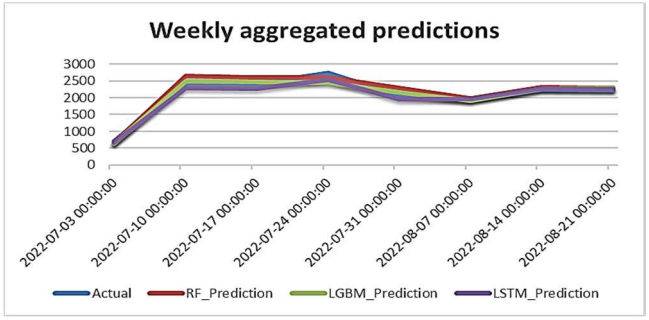



Fig. 5. Weekly aggregated predictions (Phase 2). The model follows the general trend but lacks precision.

### *C. Phase 3: The Impact of Feature Engineering.*

The final phase produced the most compelling results. The integration of engineered features caused a dramatic reduction in error for the tree-based models, leading to a convergence of performance across all three algorithms to a narrow range of 1.12-1.18 kWh.

The daily predictions in Figure 6 showcase the remarkable improvement in fidelity. The model now accurately tracks the daily consumption totals, capturing the distinct drop during weekends.



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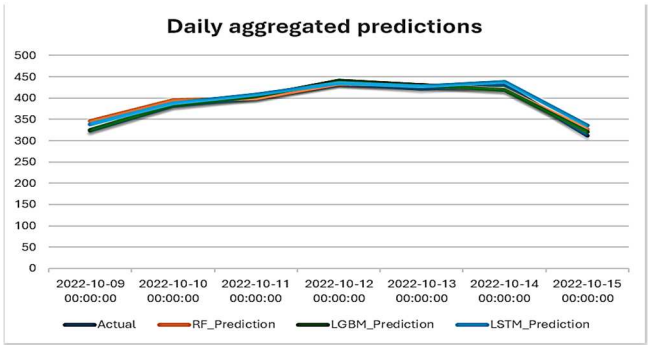



Fig. 6. Daily aggregated predictions (Phase 3). The model now exhibits high accuracy at the daily level.

The ultimate validation comes from the hourly predictions in Fig. 7. The predicted values now track the actual consumption with remarkable precision. The model accurately captures the morning ramp-up, the mid-day peak, and the evening decline, demonstrating a deep understanding of the intra-day dynamics.

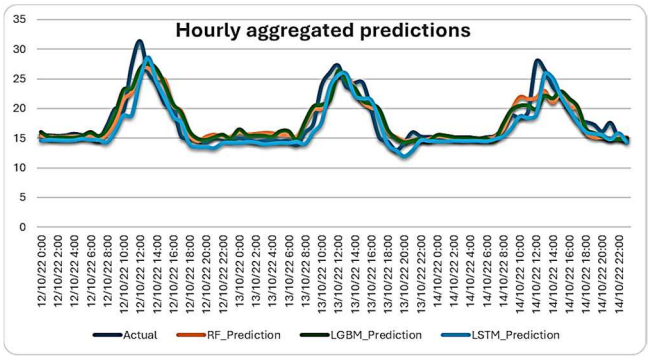



Fig. 7. Hourly predictions from the optimized LightGBM model (Phase 3). The model now exhibits a near-perfect fit, accurately capturing intra-day dynamics.

## V. Discussion

The evolutionary path of this study provides critical insights. The failure of the static models in Phase 1 highlights the danger of applying standard validation techniques to dynamic, real-world time series. The "Concept Drift" present in the data was too significant for the models to overcome without an adaptive learning strategy. The success of the sliding window in Phase 2 demonstrates that adaptability is a necessary condition for robust forecasting. By continuously retraining on recent data, the models were able to discard obsolete patterns and learn the new data distribution.

However, the leap in performance in Phase 3 proves that intelligent feature engineering is the key to sufficiency. While the sliding window taught the models *what* the new baseline was, the engineered features taught them *why* the consumption changes on an hourly basis. Lag features provided memory, rolling statistics provided a sense of immediate trend, and calendar features provided real-world context. This enrichment transformed the LightGBM model from a simple regressor into a highly accurate forecasting tool, effectively closing the gap with the more complex LSTM architecture.

## VI. Conclusion

This paper based on previous research teamwork [14, 15] successfully developed and validated a holistic methodology for short-term energy load forecasting in a university campus. We demonstrated that a combination of a dynamic, adaptive training strategy (Sliding Window) and intelligent, context-aware feature engineering is essential for achieving high accuracy. The final optimized model achieved a Mean Absolute Error of approximately 1.1 kWh, proving its reliability and practical value.

Our key finding is that the methodological process—diagnosing problems like Concept Drift and systematically enriching the data—is more critical to success than the initial choice of algorithm.

Future work will focus on three areas: (1) integrating a wider array of exogenous variables, such as humidity and solar radiation; (2) conducting systematic hyperparameter optimization to further refine the models; and (3) exploring state-of-the-art deep learning architectures, such as Transformers, to assess their potential for this application.

## Acknowledgment

The research project was supported by the European Union's Lump Sum MGA under grant agreement No 101144027 (SEBCoVE Project).

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